Nested Association Mapping (NAM) Reveals Genetic Architecture of Agronomic Traits in Quinoa

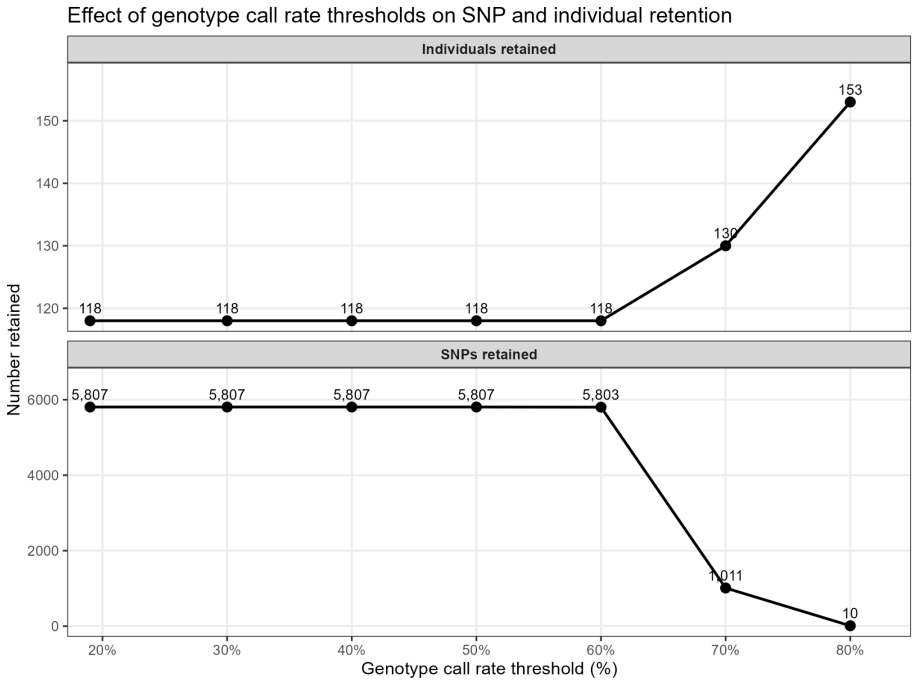
Reviewer 2

Comment: It appears that authors have attempted to revise the manuscript, but I find that Reviewer #2-Response 5 is somewhat ambiguous. Please refer to "We have revised the entire manuscript accordingly, including the Materials and Methods, Results, Discussion, Supplementary Tables and Figures, and Abstract and Conclusion, to ensure that all reported values and interpretations correspond to the 80% genotype call rate dataset." Eighty percent (genotype) call rate corresponds to 20% missing genotypes (only). But this is not what they have filtered and/or performed in the study. I would request them to please look into the issue very critically and update their understanding about the difference between 'Missing genotypes' and '(Genotype) call rate'. As a result their revised M & M section still remains confusing. I would thus recommend them to please go through that 2.3 sub-section and bring forth more clarity (before this submission is recommended for acceptance). If language is an issue, the Senior/ Corresponding author can well vet the revised draft to co-authors proficient in English.

Response:

We thank the reviewer for identifying this important methodological issue. The previous response did not adequately distinguish between genotype call rate and missing genotype rate.

To address this concern, we conducted a sensitivity analysis evaluating genotype call rate thresholds ranging from 20% to 80% and assessed their effects on the numbers of retained SNPs and individuals. The results of this analysis are now presented in Supplementary Figure 1. Based on the balance between marker retention and genotype completeness, we selected a minimum genotype call rate threshold of 60%. This filtering threshold retained 5,803 SNP markers across 118 individuals for downstream genomic analyses.



We have revised Section 2.3, Genotyping and sequence processing, to clarify the distinction between genotype call rate and missing genotype rate and to describe the revised filtering procedure. We have also updated the corresponding Results section and incorporated Supplementary Figure 1 to document the rationale for selecting the genotype call rate threshold. Importantly, the downstream genomic analyses were updated using the revised dataset, and the corresponding results, statistical thresholds, figures, tables, and interpretations were revised throughout the manuscript and supplementary materials, as applicable.

We appreciate the reviewer assessment, which helped us identify this methodological ambiguity and improve the transparency, consistency, and reproducibility of the study.